Image registration in outdoor scenes

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Abstract

What do we solve? We address the problem of advert inpainting into an outdoor scene. Existing blending techniques often produce visible artifacts due to differences in illumination and colour between foreground and background. We propose Laplace Affine Blending (LAB), an approach that combines Laplacian blending and the Poisson equation in a matrix-based framework. Our method effectively preserves structure while ensuring visually appealing blending.

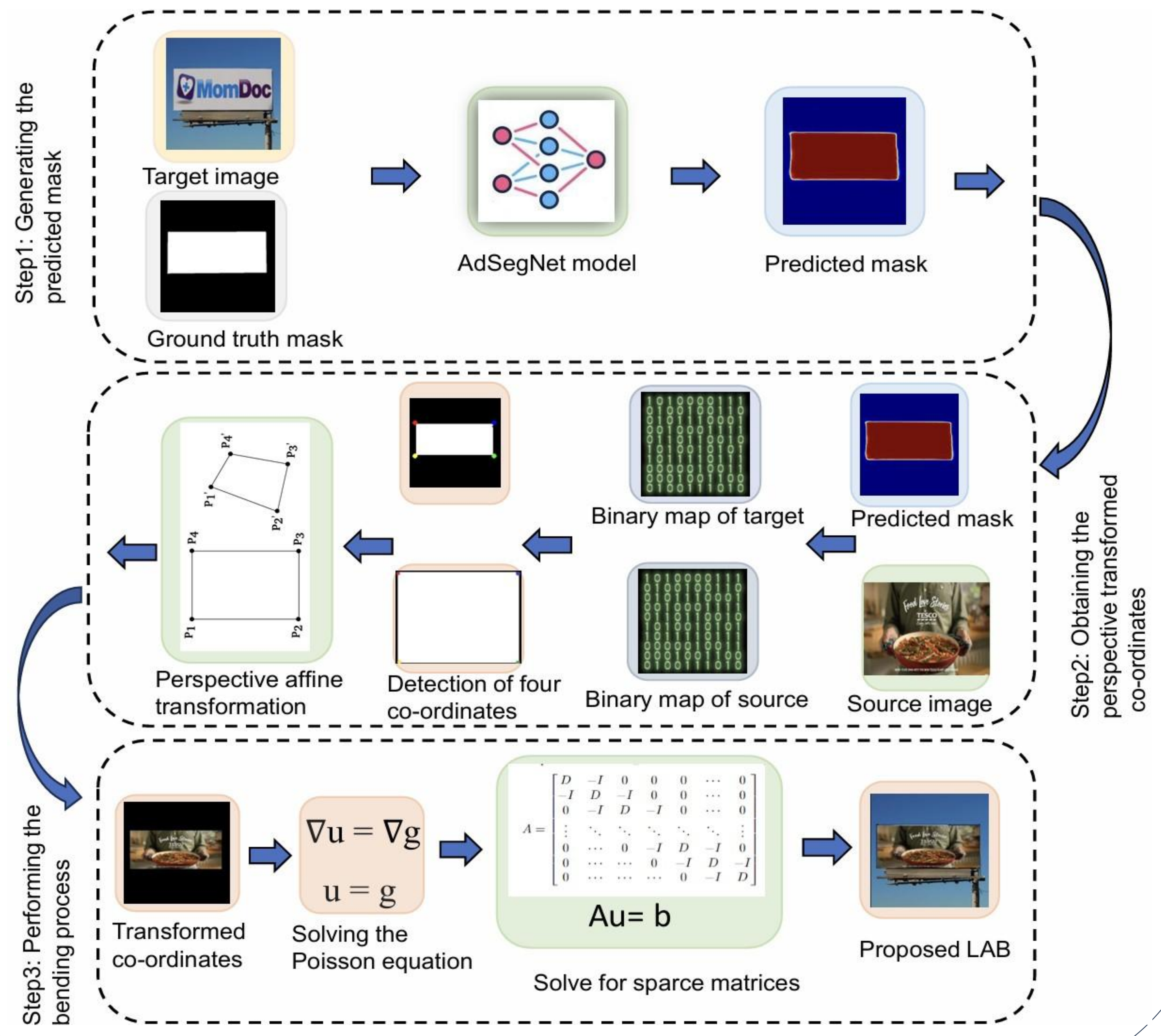
What did we find? A user study with 64 participants shows that LAB is preferred over Alpha blending, Direct-copy-paste blending, and Poisson blending in terms of overall quality and realism, and lowest perceived distortion.

Contributions

Motivation: Inserting adverts/billboards into outdoor scenes is common in advertising and visual media.

- We propose **Laplacian Affine Blending (LAB)[1]**, a matrix-based framework integrating Laplacian blending and Poisson editing.
- An extensive study shows LAB produces the most visually pleasing results with lowest distortion and variability.
- LAB is efficient and generalizes well across diverse outdoor scenes.

Proposed Methodology



Experimental Setup

Dataset: 813 outdoor background scenes [1] with diverse illumination and visibility billboard area (target image) and 5 new advert (source image).
Task: Insert advertisement using four blending methods
Goal: Produce composites that look natural to human observers.
Methods Compared:
Alpha Blending (AB)
Direct Copy-Paste Blending (DCPB)
Poisson Blending (PB)
Proposed Laplacian Affine Blending (LAB)
Participants: 64 anonymous (subjective evaluation)
Evaluation:
Preference: choose best blending method (1 out of 4)
Quality rating: 1–5 scale (1 = poor, 5 = excellent)
Distortion score: Brisque score (lower score is better)

Experimental Results

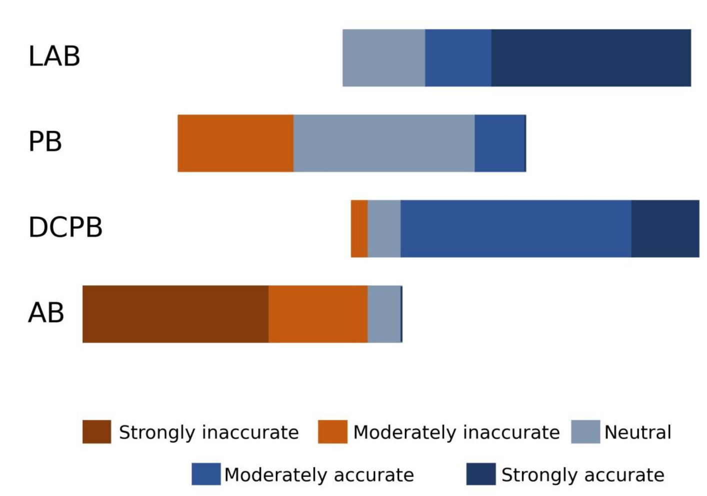
➢ Qualitative comparison

Image1

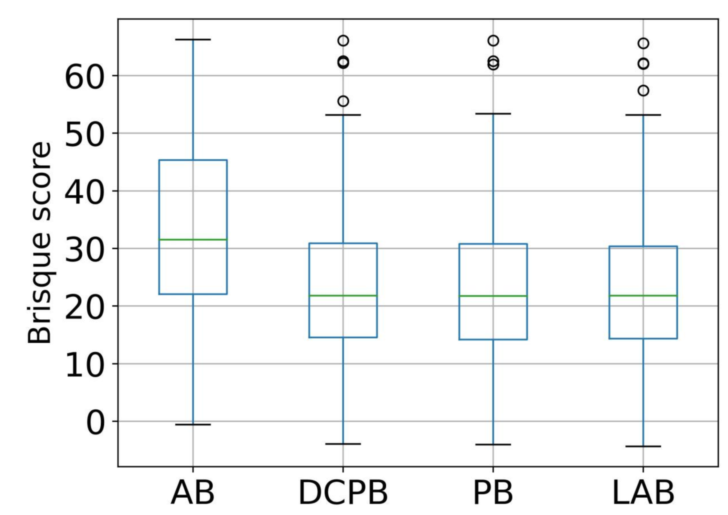
Image2

(a) Target (b) Source (c) AB (d) DCPB (e) PB (f) LAB

➢ Preference results



➢ Quality rating



➢ Distortion results: BRISQUE score

		AB	DCPB	PB	LAB
Brisque score	Image1	48.757	32.735	33.144	**32.616**
	Image2	64.341	20.942	20.742	**20.481**

Conclusion

We proposed **Laplacian Affine Blending (LAB)** for registering advertisements into outdoor scenes. By integrating Laplacian blending and Poisson editing with affine illumination adjustment, LAB effectively handles challenging lighting variations while preserving details and structure.
Future work: Extending LAB to video sequences and real-time applications.

References

[1] S. Dev, M. Hossari, M. Nicholson, K. McCabe, A. Nautiyal, C. Conran, J. Tang, W. Xu, and F. Piti´e, "The ALOS dataset for advert localization in outdoor scenes," in Proc. Eleventh International Conference on Quality of Multimedia Experience, 2019.
[2] P. P´erez, M. Gangnet, and A. Blake, "Poisson image editing," inACM SIGGRAPH 2003 Papers, 2003, pp. 313–318.
[3] G. H. Golub and C. F. V. Loan, Matrix Computations (3rd Ed.).The Johns Hopkins University Press, Baltimore, 1996.
[4] T. Remez, J. Huang, and M. Brown, "Learning to segment via cut-and-paste," in Proceedings of the European conference on computer vision(ECCV), 2018, pp. 37–52.

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[1] S. Dhang, M. Zhang, and S. Dev, "LaBlNet- An approach for seamlessly integrating new advertisement into an existing scene," IEEE Trans. on Artificial Intelligence, 2025.

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